



Logarithms

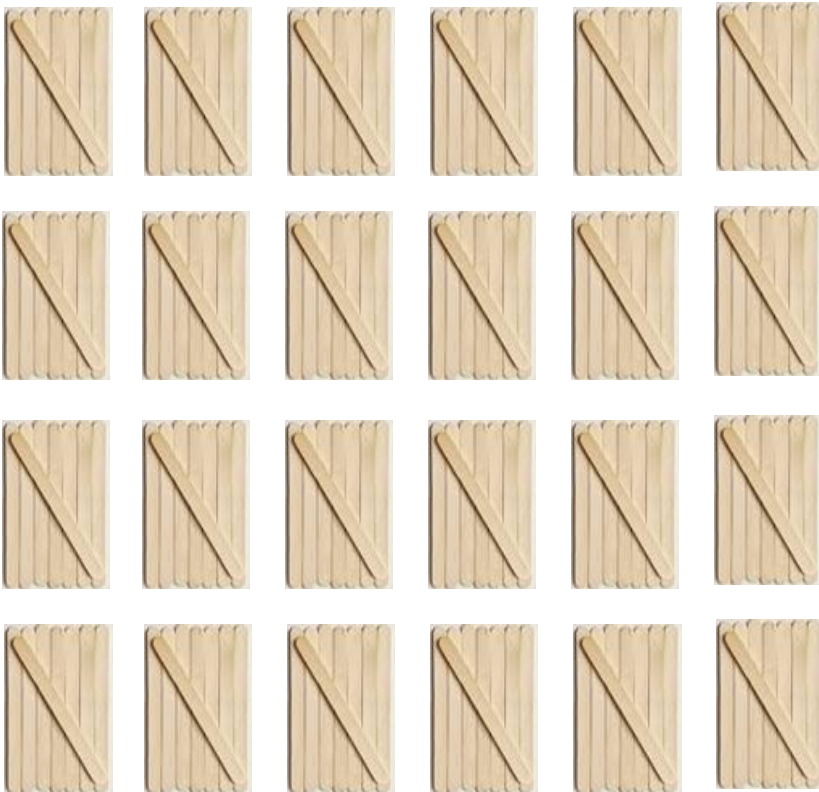
112-BOLCEW05

Concrete Experience

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What are these?
Which way is easier to count?



Why?

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Learning Outcome



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The student will solve problems using logarithms in a classroom environment, given a networked computer workstation, training aids, and appropriate references by demonstrating how to add exponentiation rules and logarithmic rules to math problems to support the understanding of Electronic Warfare Math critical to EW effects, while achieving a minimum of 70% on all Practical Exercises (PE) and end of module exam.

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LSA – Learning Step Activities



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- Exponentiation rules
- Logarithms
- Logarithmic rules
- Logarithms Practical Exercise

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Exponentiation Rules (2 of 2)



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10^n	$10 \times 10 \times \cdots \times 10$ (n tens) for any positive integer, n
10^0	1
10^{-n}	$\frac{1}{10^n}$
$10^{\frac{1}{n}}$	$\sqrt[n]{10}$ for any positive integer
$10^m \times 10^n$	10^{m+n}
$(10^m)^n$	$10^{m \cdot n}$
$\frac{10^m}{10^n}$	10^{m-n}

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Exponentiation Rules (1 of 2)



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Questions?

Logarithms



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A photograph of a whiteboard with a purple frame. A blue Expo marker lies at the top left. The equation $\log_2(8) = 3$ is written in red. Green handwritten annotations include "argument" with an arrow pointing to the 8, "base" with an arrow pointing to the 2, and "answer" with an arrow pointing to the 3.

$\log_2(8) = 3$

argument

base

answer

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Logarithms (1 of 4)



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Logarithms are the inverse operation for exponentiation.

- If the exponential form is $b^L = x$, the logarithmic form is $\log_b x = L$, where:
 - b is the base
 - L is the exponent (logarithm)
 - x is the result of raising b to the L power.
- $b^L = x$ corresponds with $\log_b x = L$

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Logarithms (2 of 4)



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Example: Find $\log_{10} 1000$

We are looking for the number k such that $10^k = 1000$

$$1000 = 10 \cdot 10 \cdot 10 = 10^3 \text{ so, } k = 3$$

$$\log_{10} 1000 = 3$$

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Logarithms (3 of 4)



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Example: Find $\log_{10} 10$

We are looking for the number k such that $10^k = 10$

$$10 = 10^1 \text{ so, } k = 1$$

$$\log_{10} 10 = 1$$

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Logarithms (4 of 4)



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Example: Find $\log_{10} 400$

Using a calculator, type press the [log] key then type in [400] and press the [=] key.

$$\cong 2.602059991$$

$$\log_{10} 400 \cong 2.60$$

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Questions?

Logarithmic Rules



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LOG RULES

$$\log_a(mn) = \log_a m + \log_a n$$

$$\log_c a = \frac{\log_b a}{\log_b c}$$

$$\log_a a = 1 \quad \log_a m^n = n \log_a m$$

$$\log_a a = 1, \log_a 1 = 0$$

$$\text{Iff } \log_a N = x \quad \log_e a = \log_e 10 \cdot \log_{10} a$$

$$\log_a \left(\frac{1}{n} \right) = -\log_a n$$

$$\text{Then } a^x = N$$

$$\text{Also } a^{\log_a N} = N$$

$$\log_a b = \frac{1}{\log_b a}$$

$$\log_{a^\beta} n^\alpha = \frac{\alpha}{\beta} \log_a n$$

$$a^{\log_c b} = b^{\log_c a} \quad (a, b, c > 0, c \neq 1)$$

$$\log_a \left(\frac{m}{n} \right) = \log_a m - \log_a n$$

$$\log_{a^m} n = \frac{1}{m} \log_a n$$

$$\log_a b \cdot \log_b a = 1$$

$$a^{\log_a m} = m$$

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Logarithmic Rules (1 of 4)



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Logarithmic Rules:

$$1. \log(N \cdot M) = \log N + \log M$$

$$2. \log N^M = M(\log N)$$

$$3. \log \frac{N}{M} = \log N - \log M$$

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Logarithmic Rules (2 of 4)



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Example: Rewrite as the log of a product.

$$\begin{aligned}\log_{11} 17x \\&= \log_{11} (17 \cdot x) \\&= \log (N \cdot M) = \log N + \log M \\&= \log_{11} 17 + \log_{11} x\end{aligned}$$

$$\begin{aligned}\log_{16} uv \\&= \log_{16} (u \cdot v) \\&= \log (N \cdot M) = \log N + \log M \\&= \log_{16} u + \log_{16} v\end{aligned}$$

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Logarithmic Rules (3 of 4)



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Example: Rewrite as the log of a quotient.

$$\begin{aligned}\log_{11} \frac{x}{17} \\&= \log \left(\frac{N}{M} \right) = \log N - \log M \\&= \log_{11} x - \log_{11} 17\end{aligned}$$

$$\begin{aligned}\log_2 \frac{3ab}{7cd} \\&= \log \left(\frac{N}{M} \right) = \log N - \log M \\&= \log_2 3ab - \log_2 7cd\end{aligned}$$

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Logarithmic Rules (4 of 4)



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Example: Rewrite as the log of a power.

$$\begin{aligned}\log_8 13^{22} \\ &= \log N^M = M(\log N) \\ &= 22 \log_8 13 \cong 27.14\end{aligned}$$

$$\begin{aligned}\log_{13} 12^{14} \\ &= \log N^M = M(\log N) \\ &= 14 \log_{13} 12 \cong 13.56\end{aligned}$$

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Questions?

Lesson Summary



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During this lesson we have discussed:

- Exponentiation rules
- Logarithms
- Logarithmic rules

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References



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- Academic Systems. “Logarithmic Properties.” in Interactive Mathematics- College Algebra. pages 248, 252, and 258. San Diego, Academic Systems Corporation, 2000, Accessed May 2019
- Marthaler, Laura and Wytko, Holly, ed., MATH for SIGINT (2007), 11-1 through 11-4

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